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ORIGINAL RESEARCH ARTICLE

The Effects of On-Field Heat Index and Altitude on Concussion Assessments and Recovery Among NCAA Athletes

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Abstract

Background Recent literature has indicated altitude may be a protective factor for concussion but it is unknown whether altitude or heat index affects recovery.

Objective To examine whether on-field heat index and altitude at the time of injury alter acute (< 48 h) concussion assessments, days-to-asymptomatic, and days-to-return-to-play in collegiate athletes following concussion.

Methods Collegiate athletes ($n = 187$; age = 19.7 ± 1.4 years; male = 70.6%) underwent baseline assessments across 30 universities and experienced a concussion in this retrospective cohort study. Altitude (m) and heat index (°C) at the time and location of injury were determined using valid online database tools. Acute concussion assessments included the Sport Concussion Assessment Tool (SCAT) symptom inventory, Balance Error Scoring System (BESS), and the Immediate Post-Concussion Assessment and Cognitive Testing (ImPACT). We used multiple linear regression models to determine whether heat index and altitude predicted each acute assessment outcome, days-to-asymptomatic, and days-to-return-to-play.

Results Collegiate athletes were concussed at a 181.1 m (range − 0.6 to 2201.9 m) median altitude and 17.8 °C (range − 6.1 to 35.6 °C) median heat index. Altitude did not predict ($p \geq 0.265$) any outcomes. Every one-degree increase in heat index reduced days-to-asymptomatic ($p = 0.047$; $R^2 = 0.06$) and days-to-return-to-play ($p = 0.006$; $R^2 = 0.09$) by 0.05 and 0.14 days, respectively. Heat index and altitude did not explain significant variance in SCAT, BESS, and ImPACT composite scores (p's = 0.20–0.922).

Conclusion Our findings suggest that on-field altitude and heat index at the time of injury do not contribute to clinically meaningful changes on acute assessments or concussion recovery. On-field altitude and heat index do not appear to significantly alter assessment outcomes or clinical recovery, suggesting that environmental factors at altitudes below < 2500 m are negligible outcomes for researchers and clinicians to consider post-concussion.

1 Introduction

Concussion is a common sports injury with growing concerns of short- and long-term health consequences in recent years [1, 2]. Large scale efforts have been made over the past two decades to examine and reduce concussion incidence in sports through equipment and rule modifications [3–5] with growing interest in identifying factors affecting concussion recovery. Factors influencing recovery such as delayed removal from play [6], early physical activity [7], and nutrition [8] have been previously explored. Despite these investigations, the role that external environment at the time of injury plays in concussion recovery is unknown.

Environmental factors consist of multiple components such as altitude, air temperature, relative humidity and heat radiation, and are strong influencers of heat illness [9].

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Key Points

Altitude was not significantly associated with assessment outcomes or clinical recovery, suggesting altitude at the time of injury is a limited factor for researchers and clinicians to consider at altitudes < 2500 m.

Increased heat index was significantly associated with decreased symptom and clinical recovery, but these finding may be of limited clinical importance.

Altitude and heat index at the time of injury are relatively unmodifiable factors, but identifying all concussion recovery influencers may further our understanding of this complex neuropathology.

Catastrophic heat illnesses occur due to excessive rise in core body temperature > 40.5 °C (104.9°F) [9], while regular participation in sports such as soccer and cross-country has been reported to raise the core body temperature above 39 °C (i.e. a 1.5 °C [2.7°F] difference) [10, 11]. Sport participation in hotter, more strenuous heat levels is also associated with dangerous increases in core body temperature [12, 13]. Though increased core body temperature alone does not cause a concussion, elevated core body temperature has been theorized to predispose individuals to greater brain injury susceptibility and severity [14–16]. Animal models have shown increased cortical contusion volume, frequency of neuronal cell death, and axonal damage following mild traumatic brain injury when core body temperature was elevated to 39 °C (102.2°F) [15, 17], suggesting increased core body temperature may contribute to worse clinical outcomes. Given regular sport participation elevates core body temperature similarly, understanding the role environmental temperature factors play in concussion recovery and clinical outcomes is important.

Recent findings in healthy individuals indicate reaction time and processing speed are impaired when exposed to hotter environments [18]. These cognitive impairments may occur in sport as well, and could place athletes at increased risk for injury if processing speed and reaction time are inhibited while on the field. Numerous studies have investigated recovery following moderate to severe traumatic brain injury, indicating more favorable outcomes when the patient's core body temperature is decreased from their resting state [19–22]. Despite these findings in moderate and severe traumatic brain injury, there is a no available research examining how the thermal environment, a strong factor contributing to core body temperature and cognitive function, may alter acute concussion presentation and recovery in humans.

Recent literature has also examined how on-field altitude at the time of injury is associated with concussion incidence with mixed results [23–25]. This body of literature faces scrutiny however as the theorized physiological changes that might prevent concussions at true high altitudes are likely not present in the aforementioned studies due to the on-field injuries occurring at altitudes well below those required to induce physiological changes in cerebral perfusion and volume [26–30]. At true high altitude (i.e. a hypobaric environment; ≥ 2500 m above sea level) [31], the human body experiences arterial hypoxemia causing impaired venous drainage, increased cerebral blood flow, increased white matter volume, and decreased cerebrospinal fluid in the brain [32–34]. These alterations may result in greater intracranial pressure [33, 35], which theoretically reduces space within the cranial vault and potentially attenuates fluid slosh [36]. Fluid dynamics indicate that increasing the pressure (i.e. decreasing space) inside a capsule will result in decreased freedom of movement, or slosh, which has been proposed and clinically implemented via jugular vein compression to theoretically cushion the brain internally to reduce damage [3, 36–39]. To truly mitigate any sloshing effect, however, empty space must be available. This is problematic given the brain and cranium both have a paucity of empty space. Though previous findings on the effect of on-field altitude on concussion incidence and the theoretical physiological processes occurring is highly debated [26–30], insight into whether altitude at the time of injury moderates acute concussion presentation and recovery may provide critical insights to the debate and offer clinical considerations post-concussion.

The purpose of our study was to determine whether on-field heat index and altitude at time of injury predicted acute concussion assessments and symptom (days from injury till being cleared to begin the gradual return to play protocol) and clinical recovery (days from injury till completing all stages of the return to play protocol) in collegiate athletes following concussion. We hypothesized (1): athletes sustaining a concussion in higher heat indexes (i.e. hotter and more humid environments) will have greater symptom scores acutely than those in lower heat indexes, but will not differ in days to asymptomatic or return to play, and (2): athletes sustaining a concussion at various on-field altitudes will not differ on any clinical assessment outcomes or days to asymptomatic or return to play. We theorized higher heat indexes would demonstrate increased symptom scores acutely due to the additional overall physical fatigue associated with physical activity in these conditions, and on-field altitude would not have any effect on clinical assessments or recovery due to mixed evidence supporting this notion [23–26].

2 Methods

This study was part of the National Collegiate Athletic Association—Department of Defense Concussion Assessment, Research and Education (CARE) Consortium, a

comprehensive study aimed at examining the effects of concussion in collegiate athletes and United States service academy members. The CARE consortium began in Fall 2014, and as of September 2018, has enrolled over 37,000 individuals with over 3,300 concussions across 30 institutions [40]. The current study only examined athletes who had completed baseline testing (Baseline) [40], experienced a concussion during enrollment in the CARE consortium with accompanying longitudinal assessments, and occurred between Fall 2014 and Fall 2018. This study is reported in accordance with strengthening the reporting of observational studies in epidemiology (STROBE) guidelines [41].

Longitudinal assessments occurred throughout concussion recovery, with the following timepoints being assessed in the present study: < 48 h, symptom recovery, and clinical recovery [40]. Symptom (defined as being cleared to begin the gradual return to play protocol) and clinical recovery (defined as completing all stages of the return to play protocol) were determined by each institution's sports medicine healthcare team with procedural guidance by the provided definitions and current consensus guidelines [42].

2.1 Participants

The initial sample of concussed collegiate athletes was 1950 and were examined for eligibility, with 187 meeting inclusion. Sample size was heavily reduced because only concussions occurring during competition, outdoor sports, and the sanctioned competitive in-season were included and eligible for accurate environmental data extraction. A flowchart outlining sample exclusion is presented in Fig. 1. Athletes were included if they suffered a documented concussion and had CARE longitudinal post-injury assessments completed [40]. Athletes were excluded if their medical records were missing key injury descriptive information (e.g. location, time of injury, injured during competitive season), reported attentional deficit disorder or attentional deficit/hyperactivity disorder, any mental health disorder, participated in a sport that performs in an indoor, climate-controlled facility, were not injured during sport activity, and were potentially experiencing Post-Concussion Syndrome (symptomatic > 4 weeks) [43]. Athletes with multiple concussions documented during CARE Consortium enrollment had only their first concussion included for analyses. The United States Army Medical Research and Materiel Command Human Research Protection Office, the University of Michigan Institutional Review Board, and each CARE consortium sites' Institutional Review Board reviewed all study procedures. All participants provided written informed consent prior to participation.

2.2 Environmental (Altitude and Thermal) Factors

Concussion injury description data were used to retrospectively determine the environmental factors at the location of each athlete's injury using Weather Underground [44–46] and Google Earth Pro [23, 47]. Specifically, we used the institution/school, sport, injury date, injury time, and injury location (home, neutral site, away) along with schedules obtained from each university's official athletic website to determine the air temperature, relative humidity, and altitude at the specific injury location on that date.

2.2.1 Air Temperature and Relative Humidity

Air temperature (°C) and relative humidity (%) at the time of the injury were retrospectively determined using Weather Underground (https://www.wunderground.com) [44], an online database operated by The Weather Channel which provides current and historically archived weather reports sampling data up to every 15 min from numerous weather stations within a specified location across the

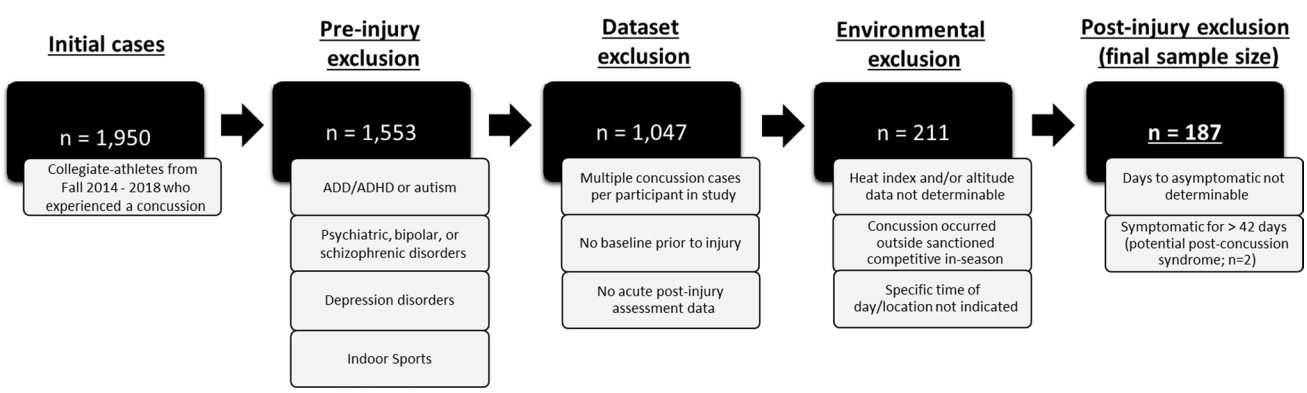

Fig. 1 Participant exclusion flowchart

United States. Weather Underground has been previously utilized in research as an accurate tool to determine retrospective reports on air temperature and relative humidity [45, 46]. Archived air temperature and relative humidity data were matched with concussion injury times to the closest timestamp available (< 1 h maximum). Two research assistants (AJS and ASG) manually matched each injury with the online Weather Underground data. Intraclass correlation coefficients (ICCs) [48] with 95% confidence intervals (CI) were computed between AJS and ASG using a sample ($n = 30$) to ensure accurate retrieval from Weather Underground, with air temperature ($ICC_{3,1} = 1.00$, 95% CI 1.00–1.00) and relative humidity ($ICC_{3,1} = 0.99$, 95% CI 0.98–1.00) data retrieval deemed reliable. Air temperature and relative humidity were used to calculate the heat index using the Rothfusz equation developed and adopted by the National Oceanic and Atmospheric Administration [49].

2.2.2 Altitude

On-field Altitude at the location of injury was determined using Google Earth Pro [47]. Research assistants operated Google Earth Pro and extracted the elevation above sea level (meters) by placing the target cursor in the center of each playing field (home, away, or neutral site locations) where the athlete suffered a concussion to determine elevation [23]. The ICC was determined for altitude among the same sample for air temperature and relative humidity ($n = 30$) with perfect agreement between raters ($ICC_{3,1} = 1.00$, 95% CI 1.00–1.00).

2.3 Concussion Assessments and Recovery

The CARE Consortium concussion baseline and post-injury assessments conducted have been previously described elsewhere [40]. In brief, baseline concussion assessments using the Sport Concussion Assessment Tool (SCAT) symptom inventory, Balance Error Scoring System (BESS), and Immediate Post-Concussion Assessment and Cognitive Testing (ImPACT) were conducted across all sites to provide individualized post-injury comparisons in the event of a suspected concussion. The same assessments were conducted across multiple time points post-injury, with the current study analyzing the < 48-h post-injury, symptom recovery, and clinical recovery time points. SCAT total symptom severity scores consisted of rating each of the 22 symptoms on a 0–6 Likert scale (0–132 score range) with a higher score indicating greater symptom severity. BESS total error scores were calculated by summing the firm and foam surface errors, with a higher score indicating greater balance impairment. ImPACT composite scores (verbal memory, visual memory, motor speed, and reaction time) were calculated as previously described [50], with higher composite scores indicating better performance with the exception of reaction time. A higher reaction time composite score indicates worse performance (i.e. more time needed to respond).

2.4 Statistical Analysis

Multiple linear regression models were utilized for each response variable (SCAT symptom severity, BESS, and ImPACT domain scores) < 48 h post-injury to examine the relationship with the continuous predictor variables (heat index [°C] and altitude [meters] at the time of injury) while controlling for previously established confounding variables (age, sex, and concussion history) [51]. On-field heat index and altitude have previously been placed into "low" and "high" categories based on median split or arbitrary cut points rather than left continuous [23–25, 45]. We chose to keep altitude and heat index as continuous variables due to unknown physiological thresholds to categorize from, and the published concerns of binning these data [26–30]. Change scores from baseline to subsequent post-injury timepoints (post-injury score–baseline score) were calculated and employed in the models for the SCAT total symptom severity score, BESS total error score, and each ImPACT composite score, with a negative change score indicating worse performance compared to baseline (positive score indicating worse performance for ImPACT reaction time composite score). Multiple linear regression models were used to determine whether on-field heat index and altitude at the time of injury predicted days to symptom recovery and clinical recovery while also controlling for age, sex, and concussion history. The altitude variable was natural logarithmically transformed to correct the positively skewed distribution. Statistical models were assessed for linearity, autocorrelation, homoscedasticity, multicollinearity, and influential datapoints, with no significant violations observed. All statistical analyses were conducted using the R Project for Statistical Programming (v 3.4.4, Murray Hills, NJ) [52] with $\alpha = 0.05$ a priori.

3 Results

A total of 187 collegiate athletes experiencing a concussion met study inclusion criteria. A subset of the 187 athletes had complete recovery data available ($n = 182$ for symptom recovery, $n = 168$ for clinical recovery), and completed symptom ($n = 121$), BESS ($n = 99$), and ImPACT ($n = 86$) assessments. Athlete demographics and concussion injury descriptive statistics are provided in Tables 1 and 2, respectively. Athletes experienced symptom recovery in 5.9 ± 5.2 days and clinical recovery in 10.7 ± 6.8 days.

Table 1 Athlete demographics ($n = 187$)

Age, years (mean, SD)	19.7	1.4
Height, cm (mean, SD)	179.4	9.9
Mass, kg (mean, SD)	83.2	18.3
Body mass index (BMI) (mean, SD)	25.6	3.7
Concussion history, yes (frequency, %)	86	46%
# Previous concussions (mean, SD)	1.5	0.9
Sex, male (frequency, %)	132	71%
Race (frequency, %)		
Caucasian	116	62%
African American	47	25%
Multiple races	15	8%
Asian	3	2%
Skipped/other	5	2%
Alaskan or native American	1	1%
Primary language (frequency, %)		
English	180	96%
Skipped/other	6	3%
Spanish	1	1%
National collegiate athletic association division (frequency, %)		
Division I	132	71%
Division II	30	16%
Division III	25	13%
Primary sport (frequency, %)		
Football	82	44%
Soccer	64	34%
Lacrosse	17	9%
Baseball	11	6%
Softball	6	3%
Cross country and track/field	5	3%
Tennis	2	1%

Collegiate athletes were concussed at a mean and standard deviation of 282.4 ± 457.1 m (media $n = 181.1$ m; range [− 0.6 to 2201.9 m]) above sea level (Table 2 and Fig. 2) and at 17.1 ± 8.3 °C (media $n = 17.8$ °C; range [− 6.1 to 35.6 °C]) heat index (Fig. 3). On-field altitude at the time of injury was not significant in any of the models ($p \geq 0.265$). Heat index at the time of injury was statistically significant within two models assessing days to symptom recovery and days to clinical recovery. When accounting for altitude and confounding variables (age, sex, concussion history), heat index predicted days to symptom and clinical recovery, with every one-degree increase in heat index shortening symptom recovery by 0.05 days ($p = 0.047$; $R^2 = 0.06$) and shortening clinical recovery by 0.14 days ($p = 0.006$; $R^2 = 0.09$). SCAT symptom severity, BESS total errors, and ImPACT composite scores were not modified by heat index or altitude in the regression models ($p \geq 0.200$). Heat index and altitude multiple regression models are summarized in Table 3.

Post-hoc multiple regression models were conducted separately for concussions occurring during home and away competitions as the on-field altitude and/or heat index may have changed quickly following injury due to travel. When account for confounding variables, days to clinical recovery ($p = 0.539$; $R^2 = 0.05$) and symptom recovery ($p = 0.834$; $R^2 = 0.03$) models were not significant nor were the variables heat index ($p \geq 0.245$) or altitude ($p \geq 0.703$) in the sample of 94 collegiate athletes concussed during home competitions. In the models for away competitions only, days to clinical recovery ($p < 0.001$; $R^2 = 0.25$) and symptom recovery ($p = 0.006$; $R^2 = 0.18$) models were statistically significant among the 89 concussed collegiate athletes. Heat index ($p = 0.006$), along with confounding

Table 2 Concussion characteristics ($n = 187$)

Injury site (frequency, %)		
Home	94	50%
Away	89	48%
Neutral site	4	2%
Immediately removed from play, yes (frequency, %)	111	59%
Time between concussion and removal, minutes (mean, SD)	57.6	94.4
Delayed symptom onset, yes (frequency, %)	60	32%
Loss of consciousness, yes (frequency, %)	17	9%
Post-traumatic amnesia, yes (frequency, %)	27	14%
Air temperature, °C (mean ± SD, median [range])	17.5 ± 7.8	18.3 (− 6.1 to 32.2)
Relative humidity, % (mean ± SD, median [range])	59.5 ± 21.6	58.0 (0 to 100)
Heat index, °C (mean, SD)	17.1 ± 8.3	17.8 (− 6.1 to 35.6)
Weather stations > 1 mile away (frequency, %)	36	19%
Distance to weather station (mean, SD)[a]	2.0 ± 1.0	
Altitude, meters (mean ± SD, median [range])	282.4 ± 457.1	181.1 (− 0.6 to 2201.9)

[a]Variable only applies to weather stations > 1 mile away

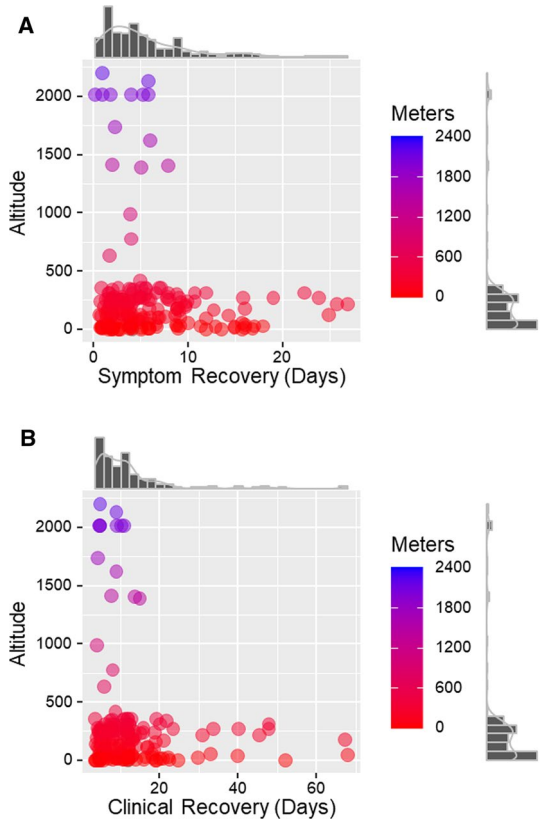

Fig. 2 Symptom (**a**) and clinical (**b**) recovery by altitude ($n = 187$) with data histogram and density plots on the respective axes

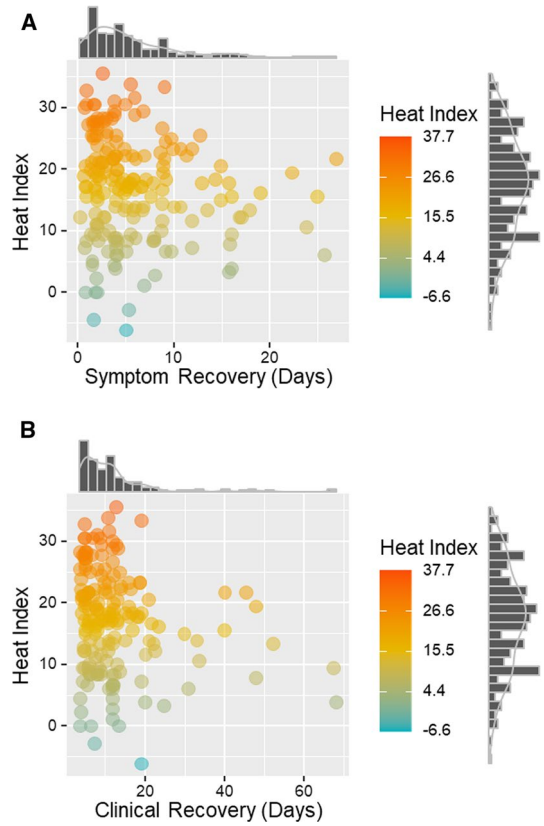

Fig. 3 Symptom (**a**) and clinical (**b**) recovery by heat index ($n = 187$) with data histogram and density plots on the respective axes

variables sex ($p = 0.008$) and concussion history (0.008), were significant predictors while altitude ($p = 0.478$) was not. Every one-degree increase in heat index shortened clinical recovery by 0.22 days, being male shortened clinical recovery by 7.12 days, and having at least one previous concussion lengthened clinical recovery by 2.88 days. Heat index ($p = 0.114$) and altitude ($p = 0.783$) were not significant predictors for symptom recovery, but confounding variables sex ($p = 0.002$) and concussion history ($p = 0.05$) were. Being male shortened the days to symptom recovery by 3.71 days, and having at least one previous concussion lengthened symptom recovery by 0.99 days.

4 Discussion

Our study provides novel insights into the role external environmental factors play in acute concussion presentation and recovery. Our findings indicate that on-field altitude below < 2500 m at the immediate time of concussion has no significant alteration on concussion recovery. Though heat index was statistically significant regarding days to symptom and clinical recovery, the clinical meaning of the small magnitude reductions in days to recover and small amount of variability accounted for by the heat index R^2 (Table 3) in the models are likely negligible.

4.1 Altitude

Our findings indicate that on-field altitude at the time of injury, specifically under the 2500 m "true high altitude" threshold, is not a significant modifier of symptom recovery, clinical recovery, or any acute clinical assessments examined. Though these are the first findings to examine concussion recovery at varying on-field altitudes representative of sport settings, previous work has examined how on-field altitude was associated with concussion incidence and have described the theoretical physiological processes [23–25]. These studies cumulatively provide mixed results regarding whether altitude altered concussion risk, which may be driven by the recently questioned methodological approaches used among studies [26–30]. There is even greater debate about the physiological factors associated with the previous studies operational use of "higher altitude". No exposures or concussions occurred at true high altitudes (2500 m above sea level) [31] in this study sample or any of the previous studies' samples, which may suggest the physiological changes were not present as well [26–30].

Table 3 Concussion recovery multiple linear regression model summary

	Heat index (°C)	Altitude (m; log scale)	Sex (male)	Concussion history (≥1)	Age (years)	Model *p*-value	R^2
Days to symptom recovery ($n=182$)	− 0.1 ± 0.1 (− 0.2, − 0.00) $p=0.047$[a]	− 0.2 ± 0.2 (− 0.7, 0.4) $p=0.535$	− 1.7 ± 0.9 (− 3.3, 0.0) $p=0.052$[a]	0.6 ± 0.4 (− 0.2, 1.4) $p=0.139$	− 0.1 ± 0.3 (− 0.6, 0.5) $p=0.789$	$p=0.045$[a]	0.06
Days to clinical recovery ($n=168$)	− 0.3 ± 0.1 (− 0.4, − 0.1) $p=0.010$[a]	− 0.6 ± 0.5 (− 1.6, 0.5) $p=0.265$	− 3.7 ± 1.8 (− 7.2, − 0.2) $p=0.040$[a]	1.4 ± 0.8 (− 0.2, 3.0) $p=0.091$	0.6 ± 0.6 (− 0.5, 1.8) $p=0.284$	$p=0.006$	0.09
< 48 h SCAT symptom severity ($n=121$)	0.0 ± 0.2 (− 0.4, 0.5) $p=0.894$	− 0.3 ± 1.2 (− 2.6, 2.0) $p=0.804$	− 3.8 ± 3.9 (− 11.5, 3.9) $p=0.329$	0.0 ± 2.0 (− 3.9, 4.0) $p=0.986$	− 0.5 ± 1.4 (− 3.2, 2.3) $p=0.739$	$p=0.922$	0.01
< 48 h BESS total error score (n = 99)	0.2 ± 0.1 (− 0.0, 0.4) $p=0.117$	− 0.6 ± 0.6 (− 1.7, 0.5) $p=0.288$	− 0.7 ± 1.8 (− 4.4, 2.9) $p=0.688$	− 0.1 ± 1.0 (− 2.1, 1.9) $p=0.927$	− 0.3 ± 0.6 (− 1.6, 0.9) $p=0.587$	$p=0.502$	0.04
< 48 h ImPACT ($n=86$)							
Verbal memory	0.1 ± 0.2 (− 0.3, 0.5) $p=0.530$	0.1 ± 1.0 (− 1.9, 2.1) $p=0.906$	6.3 ± 3.3 (− 0.3, 13.0) $p=0.062$	− 0.2 ± 1.4 (− 3.1, 2.7) $p=0.890$	− 0.5 ± 1.1 (− 2.6, 1.7) $p=0.660$	$p=0.476$	0.05
Visual memory	− 0.2 ± 0.2 (− 0.6, 0.2) $p=0.388$	− 0.3 ± 1.1 (− 2.5, 1.8) $p=0.762$	3.4 ± 3.6 (− 3.7, 10.6) $p=0.342$	1.0 ± 1.6 (− 2.0, 4.1) $p=0.505$	0.8 ± 1.2 (− 1.5, 3.1) $p=0.506$	$p=0.806$	0.03
Visual motor speed	− 0.2 ± 0.1 (− 0.2, 0.2) $p=0.819$	− 0.3 ± 0.5 (− 1.3, 0.8) $p=0.616$	0.8 ± 1.7 (− 2.5, 4.1) $p=0.636$	0.5 ± 0.7 (− 0.9, 1.9) $p=0.468$	− 0.5 ± 0.5 (− 1.5, 0.6) $p=0.391$	$p=0.816$	0.03
Reaction time[b]	3.3 ± 1.6 (0.2, 6.5) $p=0.039$[a]	− 3.1 ± 8.3 (− 19.7, 13.5) $p=0.711$	− 46.0 ± 27.5 (− 100.8, 8.8) $p=0.099$	− 4.8 ± 11.9 (− 28.4, 18.8) $p=0.688$	− 4.0 ± 8.9 (− 21.8, 13.8) $p=0.656$	$p=0.200$	0.09

All predictors reported as measure estimate ± standard error (95% confidence interval), with associated *p* value presented below

[a]Indicates statistically significant variable or model at $p \leq 0.05$

[b]Reaction time output reported in milliseconds

A previous study [23] found altitude to statistically change concussion incidence, but the authors similarly concluded that the findings lacked clinical meaning and may rather represent statistical artifact. A recent meta-analysis also found no significant effect of altitude on concussion risk in athletes [28, 53]. Our findings expand upon the meta-analytical findings by examining concussion recovery outcomes, rather than incidence, and suggest that acute clinical assessments and concussion recovery were not modified by on-field altitude. These collective findings suggest altitude across the examined NCAA sport settings is likely not a factor for clinicians or researchers to consider when assessing concussion risk, acute presentation, or recovery.

4.2 Thermal Environment

Current evidence in moderate and severe traumatic brain injury among animal and human models indicates deleterious recovery patterns and greater cortical damage when the core body temperature is elevated [14–17]. Our findings are in contrast to previous work as we observed a statistically significant relationship between heat index and recovery outcomes, with an increase in heat index reducing recovery time. Specifically, every one-degree Celsius increase in heat index significantly reduced symptom recovery and clinical recovery by 0.05 days and 0.14 days, respectively. Heat index would need to increase by 20 °C and 7 °C to reduce symptom and clinical recovery by one day, respectively. These findings are not likely to be clinically meaningful due to the small associated beta values and small explained variance (Table 3). Our findings contribute to previous research examining thermal effects on cognition and recovery after a traumatic brain injury by specifically examining a concussed cohort. However, the current findings differ from previous reports in populations and the specific assessment tools employed.

Concussions result in a complex neurometabolic cascade of events that results in immediate ionic imbalances that trigger hypermetabolic function coupled with hypoperfusion [54, 55]. Increased metabolic activity and decreased cerebral blood flow, a strong contributor to cortical thermoregulation [56], is believed to be exacerbated with higher core body temperatures. This theory is further supported by previous

research in human and animal models of moderate to severe traumatic brain injury [14–17]; however, there is no evidence in concussed populations. Our findings shed some light on the relationship between hyperthermic environments and concussion recovery in collegiate athletes, ultimately suggesting heat index at the time of injury may not have a clinically meaningful association.

Hyperthermic environments have been theorized to increase the incidence of concussions by increasing cortical tissues susceptibility to injury, via decreased cerebral blood flow with increasing core temperature [57]. A recent study examined concussion incidence among National Football League athletes and observed this theorized relationship [45]. The authors found that there was a significant increase in concussion incidence at higher air temperatures compared to colder air temperatures. The external thermal environment and core body temperature however do not appear to have any value on head impact magnitudes (linear acceleration, angular acceleration, and severity profile) as measured in collegiate football athletes [58].

These findings cumulatively suggest potential discrepancies exist between theory and research in animal models to findings in human models. Previous research has focused on understanding common intrinsic factors (e.g. sex, concussion history) and how they modify concussion recovery, but extrinsic factors such as on-field heat index and altitude have been largely ignored. Understanding how extrinsic factors relate to concussion recovery may provide advancements for future research and ultimately clinical practice. Further prospective research studies following concussion are needed among humans to fully understand the effect core body temperature and heat index have on concussion incidence and recovery.

4.3 Limitations

Though our study provides insight into external environmental factors on acute concussion presentation and recovery, there are inherent limitations. Our sample size dropped below 100 in some multivariable regression models (ImPACT and BESS), and therefore may not have thoroughly met general estimates for sample size [59]. Core body temperature was not measured in this study, but instead the external environmental factors were used as proxies to retrospectively examine if thermal stress and concussion recovery should be explored further. Another limitation is that travel after competition occurs frequently in collegiate athletics and the heat index and/or altitude individuals experienced likely changed quickly post-injury. We however attempted to control for any major environmental changes by conducting post-hoc multiple regression analyses on home and away competitions separately, with only heat index statistically influencing clinical recovery. Immediately following the competitive event that led to the injury, 59.4% were also immediately removed from play and may have had less exposure time to higher heat indexes if they were taken into a climate-controlled area. The relationship between sport time of season (e.g. football starting during relatively warmer late summer/early fall) and concussion may also be a confounding factor, as concussions may occur more frequently towards the beginning of the season in certain sports and was not controlled for in our study [60]. We also did not account for acclimatization to thermal stress or altitude as these data were not collected as part of the CARE Consortium. These factors could confound our findings; however, previous researchers have theorized that the on-field altitude and thermal stress at the time of injury are of interest [23–25]. Lastly, no athletes included in this study participated in their sport or experienced a concussion at true "high-altitude" [31]. The majority of athletic events (i.e. exposures) and concussions occurred under 500 m, and therefore our findings cannot provide insight into understanding true high altitudes role on concussion recovery. However, only a couple college institutions and athletic facilities to our knowledge are at or above 2500 m, which makes our findings more applicable to the vast majority of collegiate athletic events.

5 Conclusions

On-field altitude and heat index at the time of injury do not contribute to clinically meaningful changes in symptom or clinical recovery, nor acute clinical measures of cognition, balance, or symptom severity following concussion. Altitude was not significantly associated with assessment outcomes or clinical recovery, suggesting altitude at the time of injury is a limited factor for researchers and clinicians to consider at altitudes < 2500 m. Heat index warrants future examination to provide further clarity to the present decreases in symptom and clinical recovery that occurred at higher heat indices. While these findings were statistically significant, they may be of limited clinical importance. Altitude and heat index at the time of injury are relatively unmodifiable factors, but identifying all concussion recovery influencers may further our understanding of this complex neuropathology.

Declarations

Author Contributions All authors contributed to the study conception, study design, critical review and interpretation of findings, and drafting and/or critique of the manuscript. All authors read and approved the final manuscript. Specifically, Landon B. Lempke contributed to conceptualization, methodology, data analysis, and original and revised

manuscript drafts. Robert C. Lynall and Rachel K. Le contributed to conceptualization, analysis interpretation, and original and revised manuscript drafting. Michael McCrea and Thomas McAllister contributed to conceptualization, methodology, data analysis and manuscript critique, and funding acquisition. Julianne D. Schmidt contributed to conceptualization, analysis interpretation, original and revised manuscript drafting, and funding acquisition. The CARE Consortium Investigators contributed to critical critique and review of methodology, data analysis, and original manuscript review.

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Conflicts of Interest Landon B. Lempke, Robert C. Lynall, Rachel K. Le, Michael McCrea, Thomas McAllister and Julianne D. Schmidt declare that they have no additional conflicts of interest relevant to the content of this review.

Availability of Data and Material The CARE Consortium datasets generated and analyzed during the current study are available in the Federal Interagency Traumatic Brain Injury Research repository (https://fitbir.nih.gov/).

Research Involving Human Participants The study was performed in accordance with the standards of ethics outlined in the Declaration of Helsinki.

Informed Consent All study procedures were reviewed and approved by the University of Michigan IRB, the US Army Medical Research and Materiel Command Human Research Protection Office (HRPO), as well the local IRB at each of the performance sites. Participants provided written informed consent prior to participation.

Code Availability Not applicable.

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